

Supporting Information

Reversible β -Sheet-like Ordering and XPCS-Resolved Dynamic Arrest during Elastin-like Polypeptide Phase Separation

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1 Recombinant Synthesis and Purification of (VPAVG)₃₀

1.1 Materials

Shrimp alkaline phosphatase, PfuUltra II Fusion HS DNA polymerase, GoTaq Green Master Mix, and T4 DNA ligase with PEG4000 additive were purchased from Thermo Fisher Scientific. Exonuclease I was purchased from New England Biolabs. CircLigase ssDNA ligase was purchased from Lucigen, and KOD Hot Start DNA polymerase was purchased from Sigma-Aldrich. PCR products and plasmids were purified using Monarch PCR and DNA Cleanup kits and Monarch Plasmid Miniprep columns (New England Biolabs). DNA was recovered from agarose gels using the Monarch DNA Gel Extraction kit (New England Biolabs). Oligonucleotides and the 5'-phosphorylated synthetic template used for rolling-circle

amplification were obtained from Integrated DNA Technologies. *E. coli* BL21(DE3)-Gold cells were obtained from Stratagene.

1.2 Gene construction

The ELP gene was prepared by overlap-extension rolling-circle amplification (OERCA).[1] A single-stranded DNA template encoding five VPAVG repeats was circularized in a 20 μ L reaction containing 10 pmol template, 2 μ L of 10 $\times$ CircLigase buffer, 1 μ L of 1 mmol L⁻¹ ATP, 1 μ L of 50 mmol L⁻¹ MnCl₂, and 100 U CircLigase. The reaction was incubated for 1 h at 60 °C, heat-inactivated for 10 min at 80 °C, purified, and treated with exonuclease I for 1 h at 37 °C to remove unreacted linear template.

OERCA was performed with 300 ng circular template, 1.5 μ L each of 50 μ mol L⁻¹ primers A1 and A2 (Figure S1a), 2 μ L of 70% GC dNTP mix, 2 μ L of 25 mmol L⁻¹ MgSO₄, 10 μ L of 10 $\times$ PfuUltra II buffer, 5 μ L DMSO, and 2 μ L PfuUltra II polymerase in a total volume of 100 μ L. Cycling consisted of 2 min at 95 °C; 30 cycles of 20 s at 95 °C, 20 s at 54 °C, and 90 s at 72 °C; and a final 10 min at 72 °C. Products above approximately 1.2 kb were isolated from a 0.8% agarose gel.

Ligation-independent cloning adapters were introduced by PCR using primers B1 and B2 (Figure S1a), and the products were inserted into the pMCSG53 vector.[2, 3] Constructs were transformed into *E. coli* BL21(DE3)-Gold, screened for soluble expression, and sequence verified by Sanger sequencing. The selected construct encodes the amino-acid sequence MHHHHHSSGVDLGTE^NLYFQSNA(VPAVG)₃₀, comprising a cleavable N-terminal affinity/fusion sequence followed by 30 VPAVG repeats.

1.3 Protein expression and purification

Fresh transformants were grown in Terrific Broth containing 10 mmol L⁻¹ MgSO₄ and 300 μ g mL⁻¹ ampicillin at 37 °C to an optical density of 0.6–0.8 at 600 nm. Cultures were cooled to 18 °C, induced with 0.2 mmol L⁻¹ IPTG, and grown overnight. Cells were harvested at 5000 $\times g$, resuspended in ice-cold lysis buffer (500 mmol L⁻¹ NaCl, 5 vol% glycerol, 50 mmol L⁻¹ HEPES at pH 8.0, 20 mmol L⁻¹ imidazole, and 10 mmol L⁻¹ 2-mercaptoethanol), treated with 1 mg mL⁻¹ lysozyme for 10 min, sonicated, and clarified at 29,000 $\times g$. [4]

The supernatant was applied to 5 mL IMAC Sepharose High Performance resin, washed with lysis buffer, and eluted with lysis buffer containing 500 mmol L⁻¹ imidazole. Purification fractions were monitored by SDS-PAGE. The purified protein was exchanged into 20 mmol L⁻¹ HEPES at pH 8.0, 250 mmol L⁻¹ NaCl, and 2 mmol L⁻¹ DTT and concentrated using a 10 kDa molecular-weight-cutoff centrifugal concentrator. Protein concentration was determined by direct A_{280} measurement on a NanoDrop spectrophotometer using the sequence-specific molar extinction coefficient applied during sample preparation (calculated molecular weight 15,461 Da; $\epsilon_{280} = 1490$ M⁻¹ cm⁻¹). The final concentration was approximately 100 mg mL⁻¹; samples were stored at 4 °C, and freshly prepared protein was used for the X-ray measurements. The expression, solubility, and purity of the protein were evaluated by SDS-PAGE (Figure S1b).

2 Sample Environments and X-ray Measurements

2.1 Temperature-dependent SAXS and WAXS

Two aliquots of (VPAVG)₃₀ at approximately 100 mg mL⁻¹, nominally the same concentration used for the 8-ID-I measurements, were sealed in 2 mm quartz capillaries. The two beamtimes fell in different APS run cycles several months apart, and because the protein is purified fresh for each experiment, the 12-ID-B and 8-ID-I material came from separate preparations of the same construct. The aliquot designated

Measurement underwent more than 10 heating/cooling cycles across the thermal transition, whereas the Reference aliquot remained refrigerated at 4 °C. Both aliquots were subsequently measured at 10 °C and 30 °C during APS run 2021-1 at beamline 12-ID-B. The 13.3 keV beam had an approximate size of $100 \times 150 \mu\text{m}^2$ and an incident flux of approximately 5×10^{11} photons s^{-1} . [5, 6] SAXS was recorded with an Eiger2 9M detector at an approximately 2 m sample-to-detector distance, while WAXS was recorded with a Pilatus 300K detector at 0.42 m. The Pilatus detector was mounted above the direct-beam path, allowing the transmitted beam to continue to the SAXS detector so that SAXS and WAXS were acquired simultaneously with a 0.1 s exposure. Buffer and capillary backgrounds were subtracted, and the SAXS and WAXS profiles were stitched over their common Q interval. At each set point the capillary was held at the target temperature for at least 10 min before the exposures were started, so each profile reports the structure present after at least that long at temperature rather than during the temperature change. Because the two experiments used different sample cells, thermal paths, and thermometry, and because the time spent at 30 °C before each 12-ID-B exposure is known only as a lower bound, the 12-ID-B 30 °C profiles are not placed on the isothermal waiting-time axis of the 8-ID-I experiment; they are used here to establish the low- and high-temperature structures and their reversibility, not the kinetics of assembly.

2.2 Isothermal SA-XPCS

SA-XPCS was performed at APS 8-ID-I. Approximately 100 mg mL^{-1} (VPAVG)₃₀ in 20 mmol L^{-1} HEPES (pH 8), 250 mmol L^{-1} NaCl, and 2 mmol L^{-1} DTT was loaded into an aluminum cell containing a 3 mm-diameter, 2 mm-thick liquid chamber sealed by 0.127 mm polycarbonate windows (Figure S2). The cell was inserted into an anodized-aluminum holder in one temperature zone of the Quantum Northwest 3×3 stage. Each zone was driven by a Peltier element and a resistive heater, and a calibrated resistance-temperature detector (RTD) embedded adjacent to the Peltier stage supplied the controller feedback and readout. [7] The aluminum holder and aluminum cell provided direct conductive thermal coupling. A partially coherent 10.91 keV X-ray beam with 0.03% FWHM bandwidth and incident flux 1.2×10^{10} photons s^{-1} was focused to approximately $10 \times 10 \mu\text{m}^2$. Coherent scattering was recorded 7.8 m downstream with a Rigaku XSPA-500k photon-counting detector comprising 1024×512 pixels with a 76 μm pitch and a 320 μm -thick silicon sensor. The detector was operated in 2-bit mode at a 50 kHz frame rate; each acquisition contained 100,000 frames, corresponding to 20 μs per frame and 2 s total duration. The multi-tau delay list is built from the recorded 20 μs frame time, and the longest accessible delay is 1.638 s. [8]

For the isothermal experiment, the loaded sample was first held at 6 °C for more than 30 min. It was then heated to 20 °C at 5 °C min^{-1} , followed by a slower ramp from 20 °C to the target temperature of 30 °C at 1 °C min^{-1} . The two-stage protocol was used to reduce overshoot near the target temperature. The waiting time t_w was set to zero when the calibrated RTD readout reached 30 °C; data collected during the preceding ramps were not included in the isothermal waiting-time series. SAXS profiles were obtained by temporal and azimuthal averaging. Each reported elapsed time is a group average, not a single acquisition: consecutive 2 s acquisitions were averaged in contiguous run ranges of 63–200 acquisitions after cross-correlation outlier rejection, and the quoted t_w is the acquisition time of the first run in the range. The five analyzed groups begin at $t_w = 5039, 5969, 6595, 7221$ and 7863 s and span 7–16 min each; because f changes by less than its own group-to-group step within a group, this averaging does not materially broaden the correlation functions. Correlation functions were extracted at these elapsed times once the low- Q intensity provided sufficient photon statistics.

2.3 Experimental replication and spatial sampling

The isothermal SAXS/SA-XPCS sequence was acquired from one freshly loaded sample, and no independent sample replicate was collected. Each 2 s acquisition was recorded at a previously unexposed

spatial position, providing repeated spatial sampling within the same preparation but not independent sample replication. The five Q bins probe different length scales within the same time series and likewise are not independent replicates. Uncertainties shown for fitted quantities are one-standard-deviation uncertainties propagated from the correlation-function errors through the nonlinear fit; they do not represent between-sample variability.

3 SAXS/WAXS Reduction and Absolute-Intensity Calibration

3.1 12-ID-B SAXS/WAXS processing

The beamline-reduced `.avg` files contained Q , $I(Q)$, and the corresponding intensity uncertainty. For each aliquot and temperature, the capillary-plus-buffer profile was subtracted using a factor of 1.00 for SAXS and the empirical factor of 0.95 used in the original WAXS reduction; sample and background uncertainties were added in quadrature. No absolute WAXS intensity is used in this work—only the peak positions and widths, which are insensitive to the value of that factor. The WAXS intensity and uncertainty were then multiplied by the ratio of the mean SAXS and WAXS intensities over their common Q interval, and the two profiles were concatenated and sorted by Q . The uncertainty of this overlap scaling factor was not propagated.

For Figure 2 of the main text, the low- Q slope was obtained by an unweighted linear fit of $\log_{10} I$ against $\log_{10} Q$ over $0.012 < Q < 0.040 \text{ \AA}^{-1}$ for the Reference aliquot at 30°C . The two WAXS features were fitted independently to a Gaussian plus linear background over $0.48 < Q < 1.00$ and $1.18 < Q < 1.70 \text{ \AA}^{-1}$, respectively. The reduced profiles and scripts implementing the background subtraction, overlap scaling, Guinier analysis, power-law fit, and peak fits are available in the public repository identified in Section 7.

3.2 8-ID-I absolute-intensity calibration

For the 8-ID-I data, the differential scattering cross section was calculated as

$$I(Q) = A_s I_s(Q) - A_b I_b(Q), \quad (\text{S1})$$

where $I_s(Q)$ and $I_b(Q)$ are the temporal-mean detector intensities from the sample and buffer, per pixel and per frame. The conversion factors were

$$A_j = \frac{N_f}{F_j T_j t_{\text{exp}} L \Delta\Omega}, \quad j \in \{s, b\}, \quad (\text{S2})$$

where F_j is the incident photon flux, T_j is the transmission, $N_f = 100,000$ is the number of frames, $t_{\text{exp}} = 2 \text{ s}$ is the total exposure time, so that t_{exp}/N_f is the $20 \mu\text{s}$ frame time, $L = 2 \text{ mm}$ is the sample thickness, and $\Delta\Omega = (d_{\text{pix}}/D)^2$ is the solid angle per detector pixel.[9] Lengths are carried in millimeters throughout, so $A_j I_j$ is in mm^{-1} , and that is the quantity plotted in Figures 3a, S6b and S8: no further scaling of any kind is applied between the calibration and the figure. Absolute cross sections are more commonly quoted in cm^{-1} , which is ten times larger, since the sample thickness is the only dimensional length remaining in A_j .

Sample transmission was obtained from ion chambers immediately upstream and downstream of the sample. Across the attenuation series, the transmission of the intervening air path was independent of beam intensity within uncertainty, with a mean value of 0.8679 ± 0.0009 (Figure S3a; mean $\pm$ standard deviation). The upstream ion-chamber signal was converted to incident photon flux by calibration against a PIN diode positioned in the flight path upstream of the beamstop. The measured photon flux varied linearly with the upstream ion-chamber readout (Figure S3b). The absolute-scale SAXS evolution in Figure S8 used this transmission and flux calibration.

4 SA-XPCS Correlation Analysis

The intensity autocorrelation function was calculated using a multi-tau algorithm,

$$g_2(Q, \Delta t) = \frac{\langle I(Q, t)I(Q, t + \Delta t) \rangle_t}{\langle I(Q, t) \rangle_t^2}. \quad (\text{S3})$$

The reduction produced 27 azimuthally averaged dynamic Q bins of nominal width $1.13 \times 10^{-3} \text{ \AA}^{-1}$ spanning $0.00376\text{--}0.03307 \text{ \AA}^{-1}$. The five lowest bins, centered at $Q = 0.00376, 0.00489, 0.00602, 0.00714,$ and $0.00827 \text{ \AA}^{-1}$, are analyzed here. Both this Q restriction and the restriction to $t_w \geq 5039 \text{ s}$ are signal-to-noise limits. The scattered intensity falls by more than an order of magnitude between the lowest bin and $Q = 0.00827 \text{ \AA}^{-1}$ (Figure S8), so above that bin the correlation function is dominated by counting noise; and before $t_w = 5039 \text{ s}$ the low- Q intensity is itself too weak to resolve the second decay step. The temperature-ramp measurements of Figure S6c show the still-mobile limit of that behavior directly. The centers correspond to $2\pi/Q$ values of approximately 167, 128, 104, 88, and 76 nm.

The two-step correlation functions were parameterized as

$$g_2(Q, \Delta t) = 1 + \beta \left[f(Q) \exp \left[- \left(\frac{\Delta t}{\tau_{\text{fast}}(Q)} \right)^{p_{\text{fast}}} \right] + [1 - f(Q)] \exp \left[- \left(\frac{\Delta t}{\tau_{\text{slow}}(Q)} \right)^{p_{\text{slow}}} \right] \right]^2. \quad (\text{S4})$$

Here, $f(Q)$ is the fitted fraction of fast dynamics, $1 - f(Q)$ is the fitted fraction of slow dynamics, and $\tau_{\text{fast}}(Q)$ and $\tau_{\text{slow}}(Q)$ are effective relaxation times; these three are free at each Q . The Kohlrausch–Williams–Watts exponents p_{fast} and p_{slow} are written without a Q argument because they are shared across the five bins in the global fit, which reduces the free parameters at each elapsed time from 25 to 17. β is the instrumental speckle contrast, set by the transverse coherence of the incident beam and by the ratio of speckle size to detector pixel size; it is a property of the beamline configuration, not of the sample state, and therefore must not be allowed to vary with elapsed time or Q . It was fixed at $\beta = 0.13042$, with the baseline fixed at 1, and was measured on a static reference: a nano-porous glass standard of 10 nm nominal pore diameter (10 mm $\times$ 10 mm, 1.2 mm thick), dataset F0145, recorded at 6 °C about two hours before the isothermal series in the same beamline configuration. A static sample does not decorrelate, so $g_2(\Delta t) = 1 + \beta$. Averaging 50 repeat acquisitions at $Q = 0.02067 \text{ \AA}^{-1}$ and fitting a straight line in $\log \Delta t$ gives a flat correlation function — slope $-3.1(1.8) \times 10^{-5}$, i.e. 1.7 standard deviations from zero — with $\beta = 0.13042 \pm 0.00001$ (Figure S4a). That bin is not the best-counted one: each of the seven bins with smaller pointwise g_2 uncertainty departs further from flat, reduced χ^2 ranging from 1.9 to 17.1 against 1.02 here. It is the best-counted bin whose averaged correlation function is flat to within its own uncertainties, and so measures the instrument rather than residual structure in the standard. Every one of the 50 repeats is statistically flat at this bin (worst reduced χ^2 against a constant, 1.30), so none was rejected. Repeating the procedure bin by bin shows β to be essentially independent of Q over the range used for the sample (Figure S4b). Four of the five bins analyzed here in fact carry their own direct contrast measurement— $\beta = 0.1297, 0.1328, 0.1318$ and 0.1319 at $Q = 0.00376, 0.00602, 0.00714$ and $0.00827 \text{ \AA}^{-1}$, all within 2% of the fixed value. The fifth, $Q = 0.00489 \text{ \AA}^{-1}$, is one of four bins out of 27 rejected by the flatness cut: there the standard returns $\beta = 0.65$ with reduced $\chi^2 = 2346$, a detector or parasitic-scattering artifact rather than a contrast measurement, so no direct check is available at that Q ; the bins on either side bracket the fixed value at 0.1297 and 0.1328, giving no indication that the true contrast there differs. β is flat at approximately 0.132 out to $Q \approx 0.021 \text{ \AA}^{-1}$ and then declines steadily, reaching 0.106 in the outermost bin at $Q = 0.0331 \text{ \AA}^{-1}$. That decline is instrumental rather than a property of the standard: at 10.91 keV the 0.03% bandwidth transmitted by the Si(111) monochromator sets a finite longitudinal coherence length, and since the path-length difference across the scattering volume grows with scattering angle, contrast is progressively lost as Q increases. The reference bin used to fix β , $Q = 0.02067 \text{ \AA}^{-1}$, and all five sample bins, at $Q \leq 0.00827 \text{ \AA}^{-1}$, lie below the onset of that

falloff. Holding β fixed across Q and elapsed time is therefore supported by direct measurement over most of the analyzed range. Nor does the choice drive the result: refitting every elapsed time with β held at the extremes of the measured bin-to-bin spread, 0.1297 and 0.1328, shifts each fitted f by less than 0.01 and leaves both monotonicities intact— f decreasing with elapsed time at every Q , and increasing with Q at every elapsed time. For each elapsed time, all five Q bins were fitted jointly with p_{fast} and p_{slow} shared across Q ; f , τ_{fast} , and τ_{slow} were allowed to vary independently with Q .

Fits were performed with `scipy.optimize.least_squares` by minimizing the error-weighted residuals $[g_{2,\text{model}} - g_{2,\text{data}}]/\sigma_{g_2}$, using the pointwise g_2 uncertainties stored in the reduced HDF files. Initial values were $p_{\text{fast}} = p_{\text{slow}} = 0.5$, $\tau_{\text{fast}} = 10^{-3}$ s, $f = 0.5$, and $\tau_{\text{slow}} = 100$ s. Bounds were $0.2 \leq p_{\text{fast}}, p_{\text{slow}} \leq 3$, $10^{-6} \leq \tau_{\text{fast}} \leq 10$ s, $0 \leq f \leq 1$, and $1 \leq \tau_{\text{slow}} \leq 10^4$ s. The maximum number of function evaluations was 40,000. Global fits with reduced $\chi^2 \geq 10$ were excluded from the plotted fit-parameter trends; no analyzed elapsed time meets that condition—the five give reduced $\chi^2 = 0.78$ –1.89—so the cut removes nothing. The covariance matrix was estimated as the Moore–Penrose pseudo-inverse of $J^T J$, with J the Jacobian of the error-weighted residuals at the solution, so that a parameter refining to a bound (see above) does not render the uncertainties of the remaining, well-constrained parameters undefined; reported one-standard-deviation parameter uncertainties are the square roots of its diagonal elements. These are the uncertainties plotted in Figure 3c, and the significance of the Q dependence of f is formed from them: the difference between two bins is compared with $\sigma = \sqrt{\sigma_i^2 + \sigma_j^2}$. For adjacent bins at $t_w = 5039$ s the four differences are 2.0, 2.2, 2.9 and 1.8σ ; at the four later times the smallest adjacent difference is 8.6, 11.4, 13.3 and 10.8σ . Taken end to end, from the lowest to the highest Q bin, the difference is 9.0, 57, 62, 87 and 39σ at the five elapsed times. The increase of f with Q is therefore resolved at every elapsed time, marginally so between the closest pair at the earliest time and overwhelmingly thereafter. Power-law relations $\tau = Aq^\gamma$ in Figure S9 were obtained by weighted linear regression of $\ln \tau$ against $\ln Q$, with $\sigma_{\ln \tau} = \sigma_\tau / \tau$.

The slow component remains only partially relaxed during the 2 s acquisition and frequently yields $\tau_{\text{slow}} > 2$ s. It is therefore an effective extrapolation rather than a fully sampled terminal relaxation time. Conclusions regarding arrest are based primarily on directly observed features: the decreasing amplitude of the first decay, the rising plateau, and the systematic decrease in f , equivalently the increase in the slow fraction $1 - f$. The monotonic rise of $1 - f$ indicates that a statistically increasing fraction of the scattering ensemble is dynamically arrested within the experimental window. XPCS does not determine where those regions are located, and $1 - f$ is not a direct geometric volume fraction because scattering contributions are weighted by structure and contrast. Because the measurement does not reach infinite delay, $1 - f$ is also not identified with the formal infinite-time nonergodicity factor. Figure S9 reports all auxiliary fit parameters; Figure S10 shows the measured correlation functions and fits at the four additional Q values: the same two-step shape, and the same monotonic loss of the fast step with elapsed time, are present at every measured Q , so the trend of Figure 3c is a property of the whole 76–167 nm range and not of a single bin.

5 Radiation-Damage Mitigation and Controls

A Cartesian grid of 100×100 sample positions was defined in the aluminum cell, with adjacent points separated by at least one beam diameter. Every 2 s acquisition was collected at a previously unexposed position, preventing cumulative dose from tracking the elapsed time of the isothermal experiment.

An independent dose-rate test was performed after the APS Upgrade using a 20 mg mL^{-1} (VPAVG)₃₀ sample. The sample was held above the transition, at a stage-RTD readout of 33.0°C for every acquisition, so the control was recorded on an assembled rather than a dispersed state. Attenuation selected incident fluxes from 2.7×10^{11} to 4.9×10^{11} photons s^{-1} , approximately 23–41 times the flux used for the main

experiment. The SAXS line shapes and g_2 decays showed no systematic flux dependence (Figure S5). At this lower concentration the correlation function still decays essentially to the baseline within the accessible delay range, so the control samples an assembled but still mobile state and does not reproduce the arrested state of the primary experiment. Together with the lower protein concentration and the different beam configuration, this means the control tests for gross flux-driven artifacts rather than reproducing the complete main-measurement condition.

6 Supporting Structural and Dynamical Results

6.1 Macroscopic reversibility

Video S1 provides a direct visual demonstration of thermal reversibility. Brief warming of a capillary containing (VPAVG)₃₀ produces turbidity within approximately 5 s, indicating structures large enough to scatter visible light. After removal of the heat source, the sample returns to optical transparency in approximately 60 s. This recovery is consistent with the overlap of the cycled Measurement and refrigerated Reference SAXS/WAXS profiles at 10 °C. Optical transparency does not specify molecular structure, but independently confirms dissolution of the mesoscale assembly on cooling.

6.2 Thermal-cycling repeatability of the SA-XPCS sample

The reversibility comparison of Figure 2 sets two separate 12-ID-B aliquots against each other. A more direct test was made at 8-ID-I, where a single aliquot was taken through seven consecutive thermal cycles and compared with itself. This aliquot is not the loading used for the isothermal experiment of Figure 3 of the main text: both are aliquots of the same stock, measured two days apart in the same cell position of the same nine-position temperature-controlled holder during the same beamtime, with the cell reloaded in between. The cycling sequence was: held at 6 °C, ramped to 34 °C at 1.01 °C min⁻¹ over 270 consecutive 2 s acquisitions, then returned to 6 °C, over a total of 5.8 h (Figure S6a). Sample temperature was read from the calibrated RTD of the stage zone holding the cell, once per acquisition. Every group averaged in Figure S6—the seven 6 °C groups, the two high-temperature windows of each ramp, and the buffer—is first passed through the same two automatic outlier cuts used for the isothermal groups of Figure 3: the cross-correlation rejection at the same 5th-percentile cutoff, followed by the low- Q intensity cut described below. Between 32 and 41 acquisitions of each 50-acquisition 6 °C group survive both, and those are averaged. Every group in this paper is reduced by the same two cuts on the same terms; none is reduced by hand. Because no acquisitions were taken while the sample cooled, the temperature over those intervals was not recorded at all; in Figure S6a it is shown dashed and is a projection from the fixed control protocol (cool at 10 °C min⁻¹ to 6 °C, then hold), not a measurement. Every solid segment in that panel is measured, and the open symbols mark where each group plotted in (b) and (c) was taken.

Because every curve comes from one and the same material, any difference between cycles can only be irreversibility. Figure S6b shows the buffer-subtracted absolute-scale SAXS profile at three states sampled in every cycle: 6 °C (the retained acquisitions of all 50 averaged), and two windows near the top of each ramp, acquisitions 241–250 (31.9 °C) and 261–270 (33.8 °C). Averaged over $0.004 < Q < 0.008 \text{ \AA}^{-1}$, the seven cycles agree to a relative standard deviation of 9% at 6 °C and 5% at 33.8 °C, with no systematic drift from the first cycle to the seventh. The 6 °C profile still rises toward low Q rather than reaching the flat Guinier plateau expected of a fully dispersed solution. This is not mis-subtracted solvent: the buffer contributes only about 35% of the 6 °C signal at the lowest Q , so removing the rise by rescaling it would subtract more than three quarters of the low- Q signal and degrade the cycle-to-cycle agreement from 9% to 23%, while changing the 33.8 °C curves by less than 1%. The rise is therefore left in. Its origin is identified below. The spread is larger, 33%, at 31.9 °C, and that is a sampling difference rather

than a difference between cycles. Because the two high-temperature windows are taken on a ramp, each cycle crosses them at a slightly different temperature: acquisitions 241–250 fall between 31.62 °C and 32.48 °C across the seven cycles. That temperature lies on the steep part of the transition, where the intensity is changing fast enough for a few tenths of a degree to matter. The correction is a single one-parameter regression over the seven cycles, not a fit to each cycle separately: the band-averaged $\ln I$ of a cycle is regressed on the mean temperature of that cycle’s window, which gives $d \ln I/dT = 0.95 \text{ } ^\circ\text{C}^{-1}$ with $r = 0.974$, and the relative standard deviation is then recomputed on the residuals about that line rather than on the raw intensities. It falls from 32.6% to 7.1%. The same regression at 33.8 °C, where the windows span only 33.53–34.00 °C and the transition has flattened to $d \ln I/dT = 0.28 \text{ } ^\circ\text{C}^{-1}$, takes 4.8% to 1.4%. The 33% is therefore a statement about where on the ramp each cycle was sampled, not about how well the cycles repeat. The cycles therefore repeat to better than 10% at every state once the temperature at which each was sampled is accounted for.

The 6 °C profile of Figure S6b also falls off more steeply toward low Q than the 6 °C reference plotted in Figure 3a of the main text: log-log slopes of -1.0 and -0.5 respectively below $Q = 0.01 \text{ } \text{\AA}^{-1}$. The two are reduced identically—per-acquisition absolute scaling, the same outlier rejection, and the averaged buffer subtracted with no empirical coefficient—and over $0.005 < Q < 0.032 \text{ } \text{\AA}^{-1}$ their shapes agree to within 22%, so the difference is confined to the lowest Q and lies in the acquisitions themselves. The per-acquisition distribution identifies it. Within a single 6 °C group of the cycling aliquot, and taking every acquisition as recorded, the intensity at $Q = 0.0035 \text{ } \text{\AA}^{-1}$ is strongly right-skewed: relative standard deviations of 51% and 96% for two of the groups, with maxima 3.4 and 8.0 times the group median (skewness $+2.2$ and $+3.1$). The same acquisitions at $Q = 0.03 \text{ } \text{\AA}^{-1}$ are symmetric with relative standard deviations near 9%, that is, ordinary counting statistics. A long positive tail confined to low Q is the signature of a large object crossing the $10 \times 10 \text{ } \mu\text{m}^2$ beam in a minority of acquisitions. The isothermal aliquot of Figure 3 does not show it (relative standard deviation 5.8%, maximum 1.17 times the median). The cross-correlation rejection does not remove these acquisitions, because it compares curve shape over the whole Q range and a curve that is anomalous in three low- Q bins alone remains close to the group mean by that measure.

A second, robust cut removes them. For every group in the paper, the mean of $I(Q)$ over $0.004 < Q < 0.008 \text{ } \text{\AA}^{-1}$ —the same band this section quotes its repeatability in—is formed for each acquisition, and an acquisition is rejected when that mean lies more than three robust standard deviations above the group median, the scale being $1.4826 \times$ the median absolute deviation and both the median and the deviation recomputed from the survivors until the set stops changing. The test is one-sided because a transient scatterer only adds intensity. It is applied to every group on identical terms, with no per-group list, and it removes 35 of the 1708 acquisitions that survive the cross-correlation cut, that is 2.0%. In the seven 6 °C groups it removes 4, 2, 1, 11, 4, 1 and 1 acquisitions and from the two buffer groups a further 2 and 7, while removing none at all from any of the fourteen ten-acquisition high-temperature windows, consistent with the skew being confined to the cold, low- Q state. Outside this figure the cut is nearly inert: it removes two acquisitions of the isothermal series, which shifts one absolute-scale coefficient by 0.03% and leaves every fitted correlation-function parameter of Figures 3, S9 and S10 unchanged. The effect on this figure is reported rather than absorbed: the fourth cycle, the group with the 96% spread, averaged 31% above the mean of the other six over $0.004 < Q < 0.008 \text{ } \text{\AA}^{-1}$ before the cut and 7% below it after, and the cycle-to-cycle relative standard deviation at 6 °C falls from 13% to 9%. What remains is a rise over the last two cycles rather than one anomalous group. The threshold is not tuned: the same acquisitions are selected for any threshold between 2.5 and 3, and for bands of $0.0032 < Q < 0.006$ or $0.003 < Q < 0.005 \text{ } \text{\AA}^{-1}$. No conclusion drawn from this figure depends on the cut, which rests on the ratio between cycles and not on their common level.

A difference between the two loadings survives all of this and is a property of the material rather than of a few acquisitions: the cycling loading has $I(0.004)/I(0.02) = 4.0$ against 2.7 for the isothermal loading,

and transmits 0.305 against 0.333. It was the less completely dissolved of the two.

Figure S6c shows the corresponding correlation functions at the lowest Q bin. At 33.8 °C g_2 is flat at $1 + \beta$ across the whole accessible delay range in every cycle, the signature of an arrested state, and independently confirms the contrast measured on the static reference. At 31.9 °C the correlation decays essentially to the baseline within the same window, so the assembly is still mobile there. This is also the direct evidence behind the elapsed-time restriction applied in the main text: while the sample is still mobile the low- Q correlation completes a single decay inside the accessible window, so there is no second, slower step there to resolve. No correlation function is plotted for the 6 °C state. There the sample is a dispersed solution and the coherent signal at the lowest Q is too weak for any contrast to be resolved above the noise, so the measured g_2 carries no usable information; the two high-temperature states are shown precisely because they do. Both the contrast and the baseline are separately visible in the 31.9 °C curve—the condition under which it can be interpreted at all. Even at that temperature the per-acquisition statistics are poor: the curves in Figure S6c are averages over ten consecutive acquisitions, and a single 2 s acquisition does not resolve the plateau.

6.3 Low-temperature unassembled state

The Reference profile at 10 °C contains a Guinier regime satisfying $Q_{\max}R_g = 1.246$ (Figure S7). A linear fit of $\ln I(Q)$ against Q^2 gives

$$R_g = 22.65 \pm 0.50 \text{ \AA}. \quad (\text{S5})$$

The fitted size is molecular in scale and is consistent with a dispersed, unassembled low-temperature state rather than persistent mesoscale aggregates. Because the protein concentration is high, this value is reported as an apparent R_g ; weak intermolecular structure-factor contributions cannot be excluded. Recovery of the same low-temperature profile after repeated cycling establishes that the large high-temperature assembly is no longer detectable after cooling.

6.4 Isothermal SAXS evolution at 30 °C

Figure S8 shows the full SAXS sequence. At $t_w = 0$, the profile is close to the 6 °C reference. The low- Q intensity then increases continuously and ultimately by more than two orders of magnitude, documenting progressive growth of mesoscale density contrast. The profile remains broad throughout, consistent with hierarchical network development rather than selection of a narrow domain spacing. The most pronounced increase occurs between $t_w = 4088$ and 5039 s, after which the intensity continues to grow through 7863 s. This growth is what eventually brings the low- Q coherent intensity above the level at which a two-step correlation function can be resolved, and is therefore the origin of the $t_w \geq 5039$ s restriction. The correlation analysis in Figure 3 of the main text uses $t_w = 5039$ –7863 s; earlier scattering was insufficient for reliable two-step correlations at the 20 μ s frame time. Over the $0.012 < Q < 0.033 \text{ \AA}^{-1}$ interval common to both beamlines—narrower at the top end than the $0.012 < Q < 0.040 \text{ \AA}^{-1}$ range behind the slopes quoted in the main text—the 12-ID-B 30 °C profiles give power-law slopes of -3.09 (Reference) and -3.07 (Measurement), whereas the isothermal 8-ID-I series steepens from -2.45 at $t_w = 5039$ s to between -2.64 and -2.68 over 5969–7863 s. The isothermal series is thus still short of the structure reached by the 12-ID-B capillaries, and the two datasets are complementary rather than replicate measurements of one state.

6.5 Length-scale-dependent relaxation parameters

Figure S9 shows parameters from the global fits. Both shared stretching exponents remain below unity: $p_{\text{fast}} \approx 0.36$ –0.70 and $p_{\text{slow}} \approx 0.26$ –0.57. These stretched decays indicate broad relaxation-time

distributions, consistent with a heterogeneous, nonequilibrium network.[10, 11]

The effective fast relaxation occurs on approximately 1×10^{-4} – 2×10^{-3} s and generally becomes faster with increasing Q . At each elapsed time, the Q dependence was parameterized as $\tau(Q) \propto Q^\gamma$. The fast-mode exponent is close to -2 at every analyzed elapsed time, $\gamma_{\text{fast}} = -2.0$ to -2.4 . At the final elapsed time the fast amplitude at the lowest Q bin refines to $f = 0.000 \pm 0.003$, i.e. no fast mode is detected there and τ_{fast} is undefined; that bin is therefore excluded from the fast-mode panel of Figure S9 and from the corresponding power-law fit, which is why the final γ_{fast} carries a larger uncertainty. The fitted slow component also decreases with Q , but its tens-to-thousands-of-seconds values exceed the acquisition window; neither τ_{slow} nor γ_{slow} is interpreted as a directly measured terminal relaxation law. The robust arrest measure remains the monotonic decrease in the fitted fast fraction f —equivalently, the increase in the fitted slow fraction $1 - f$ —across all five Q values.

7 Code and Reduced-Data Availability

The reduced datasets and all analysis and figure-generation code used for this manuscript are publicly available at <https://github.com/qzhang234/PubCode-2021-IDPP>. The repository includes the reduced 12-ID-B SAXS/WAXS data, the averaged 8-ID-I XPCS HDF files for every group plotted, the per-acquisition temperature record of the thermal-cycling sequence, the individual correlation functions of the speckle-contrast standard, radiation-control data, absolute-intensity calibration inputs, and the scripts used to generate the main-text and Supporting Information figures. Every data-derived figure can be regenerated from the repository alone, and two further scripts, one per beamline, record how the committed reduced files—including both outlier-rejection passes, applied to every group on the same terms—were produced from the raw beamline results. The repository README maps each script and output file to the corresponding manuscript figure. Raw detector-frame datasets are not included in the repository and are available from the corresponding authors upon reasonable request.

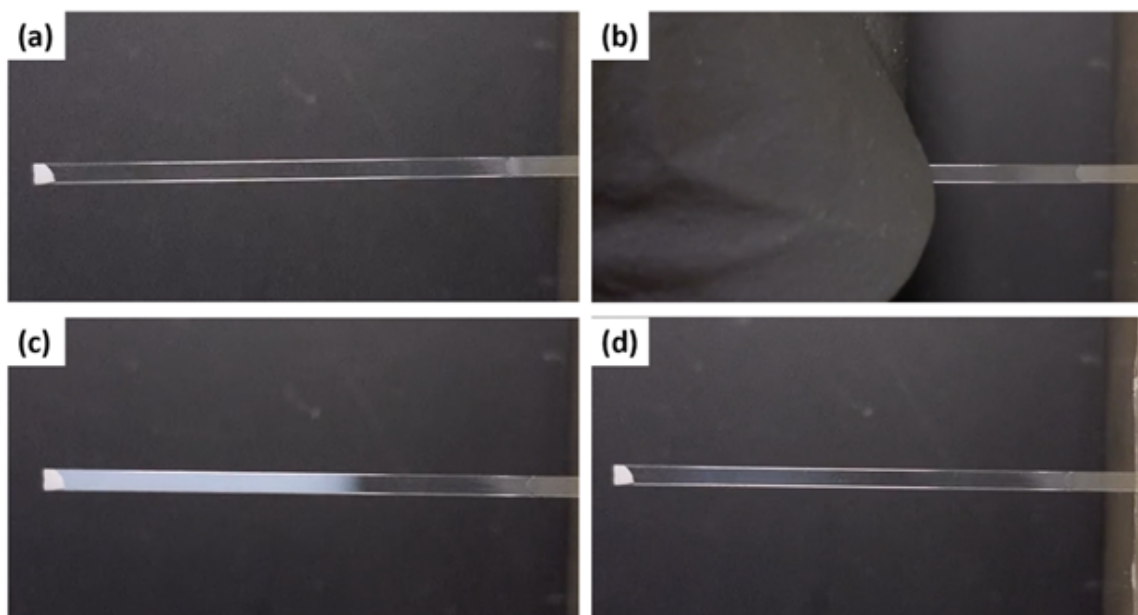
8 AI-Assisted Development

The authors used Claude Sonnet 5 and Opus 5 to assist with the manuscript preparation and code development. The Claude models were accessed through Argo, Argonne National Laboratory’s internal generative AI interface, which provides Laboratory staff with access to large language models from multiple vendors, including Anthropic, OpenAI, and Google, for use in Laboratory research and operations.

The Python scripts read the experimental and reduced SAXS/WAXS and XPCS datasets; perform background subtraction, SAXS/WAXS stitching, absolute-intensity calibration, Guinier and peak fitting, global two-mode XPCS fitting, uncertainty propagation, and Q -dependent scaling analysis; and generate the data-derived plots used in the manuscript and Supporting Information. All of that code, and the reduced data it operates on, are publicly available at <https://github.com/qzhang234/PubCode-2021-IDPP> (Section 7), so every reported number and data-derived figure can be regenerated and checked independently.

Qingteng Zhang reviewed the code, reran the analyses, and confirmed the reported results; all analysis choices, scientific interpretations, and manuscript content were determined by the authors, who take full responsibility for the publication.

Supporting Video



Video S1. Reversible thermal transition in (VPAVG)₃₀. (a) A thin quartz capillary containing (VPAVG)₃₀ is held in a fixture. (b) The capillary is warmed for approximately 5 s by contact through nitrile gloves. (c) The initially transparent solution becomes visibly turbid. (d) The sample returns to transparency approximately 60 s after removal of the heat source. The complete video can be viewed or downloaded at <https://an1.app.box.com/s/lsukldjnr45nhfrkobij sar1tqhba7jw>.

(a)

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A1 (forward)
5'-GTTCCAGCTGTTGGTG

A2 (reverse)
5'-ACCTACTGCTGGTACG

B1 (forward)
5'-TACTTCCAATCCAATGCCGTTCCAGCTGTTGGTGTGCCG

B2 (reverse)
5'-TTATCCAATTCCAATGTTAACCTACTGCTGGTACGCCGAC

VPAVG30 OERCA single-stranded template
/5Phos/GTTCCAGCTGTTGGTGTGCCGGCCGTGGGCGTGCCTGC
GGTGGGCGTCCCGGCGTCCGGCTACCAGCAGTAGGT

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(b)

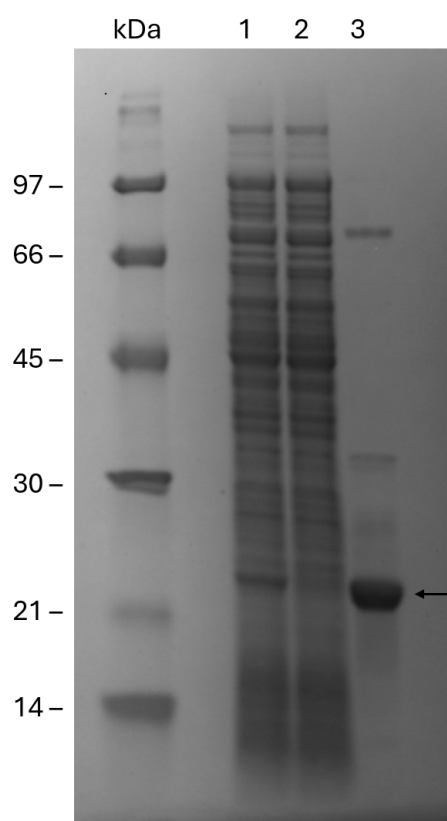


Figure S1: Construction and expression of the (VPAVG)₃₀ gene. (a) Oligonucleotide primers and the 5'-phosphorylated single-stranded OERCA template used to construct the ELP gene. (b) SDS-PAGE analysis of expression and purification. Samples were resolved by SDS-PAGE and stained to evaluate expression, solubility, and purity of the heterologously expressed protein. The molecular-weight marker lane is at the left, with prominent bands labeled in kilodaltons (kDa). Lanes 1–3 are the total soluble fraction, the flow-through fraction after IMAC, and the eluted fraction, respectively. The arrow marks the eluted (VPAVG)₃₀ construct, whose apparent mobility is lower than that expected from its calculated mass of 15,461 Da.

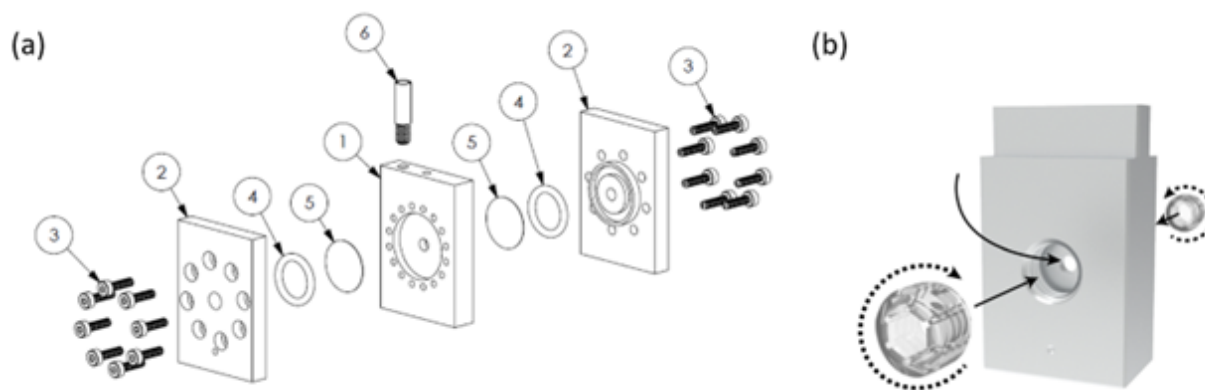


Figure S2: Sample cells used for liquid X-ray measurements. (a) Exploded view of the aluminum cell used for the 8-ID-I isothermal SA-XPCS experiment. The cylindrical chamber is 3 mm in diameter and 2 mm thick along the X-ray direction; 0.127 mm polycarbonate windows are compressed by Viton O-rings and bolted caps. During measurement, this cell was inserted into the anodized-aluminum holder of the Peltier-controlled stage described in Section 2.2; both the holder and cell were aluminum. (b) Screw-cap cell developed separately for automated handling; the dashed circles magnify the window seat. Only the bolted aluminum cell in panel (a) was used for measurements reported in the main text. The circled numerals in (a) identify the components: (1) the aluminum cell body, which contains the sample chamber; (2) the aluminum cap plates; (3) M2 screws, 8 mm long, which clamp the caps against the body; (4) Viton O-rings; (5) window disks cut from Kapton or polycarbonate sheet with a hammer-driven punch, polycarbonate being used for the measurements reported here; and (6) the handle used to withdraw the cell from the Quantum Northwest stage. Labels (2)–(5) appear on both sides of the body because the assembly is symmetric about the sample chamber.

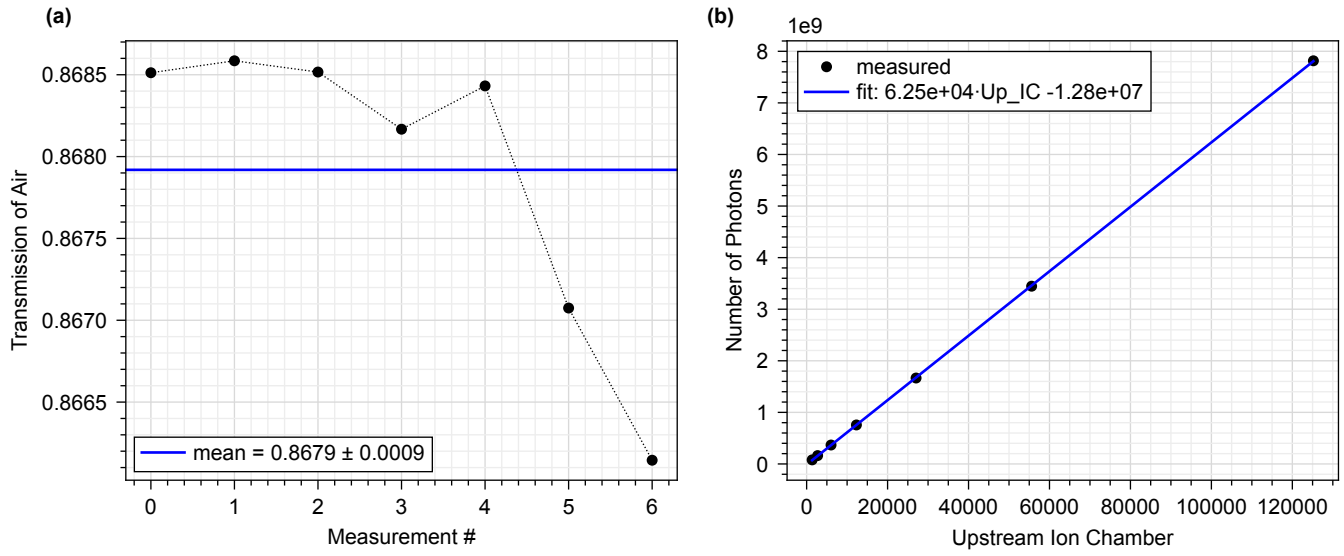


Figure S3: Transmission and photon-flux calibration for conversion to absolute scattering cross section. (a) Air-path transmission across seven measurements in the attenuation series; the line marks the mean, 0.8679 ± 0.0009 (standard deviation). (b) Photon counts recorded by a calibrated PIN diode as a function of the upstream ion-chamber readout (both in the scaled detector units of the calibration series); the line is the linear calibration fit, from which the incident flux follows as $F = (A \cdot \text{Up_IC})/t_{\text{exp}} + B$.

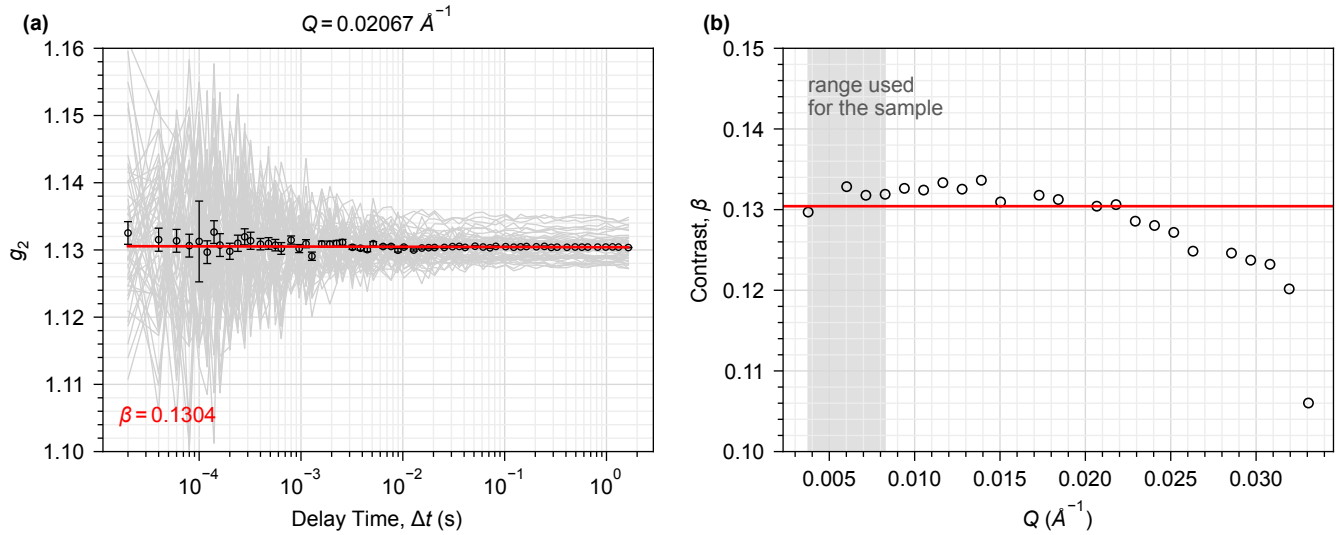


Figure S4: Measurement of the instrumental speckle contrast β on a static reference. (a) Correlation functions of a 10 nm nano-porous glass standard at $Q = 0.02067 \text{ \AA}^{-1}$, the best-counted bin whose averaged correlation function is flat to within its uncertainties: 50 individual repeat acquisitions (grey) and their average (black, with the measured pointwise uncertainties). The red line is a straight-line fit in $\log \Delta t$; it is flat to 1.7 standard deviations with reduced $\chi^2 = 1.02$, as a static sample requires, and its level gives $\beta = 0.13042 \pm 0.00001$. (b) The same procedure applied bin by bin. β is essentially independent of Q across the shaded range used for the sample; the red line is the fixed value from (a). Above $Q \approx 0.021 \text{ \AA}^{-1}$, outside that range, β falls off as the finite longitudinal coherence length set by the Si(111) monochromator begins to limit the contrast. Four of the five bins used for the sample return $\beta = 0.1297, 0.1328, 0.1318$ and 0.1319 , all within 2% of it. Four bins out of 27 whose averaged correlation function is not flat (reduced $\chi^2 > 5$, detector and parasitic-scattering artifacts) are omitted, one of them at $Q = 0.00489 \text{ \AA}^{-1}$ inside the shaded range.

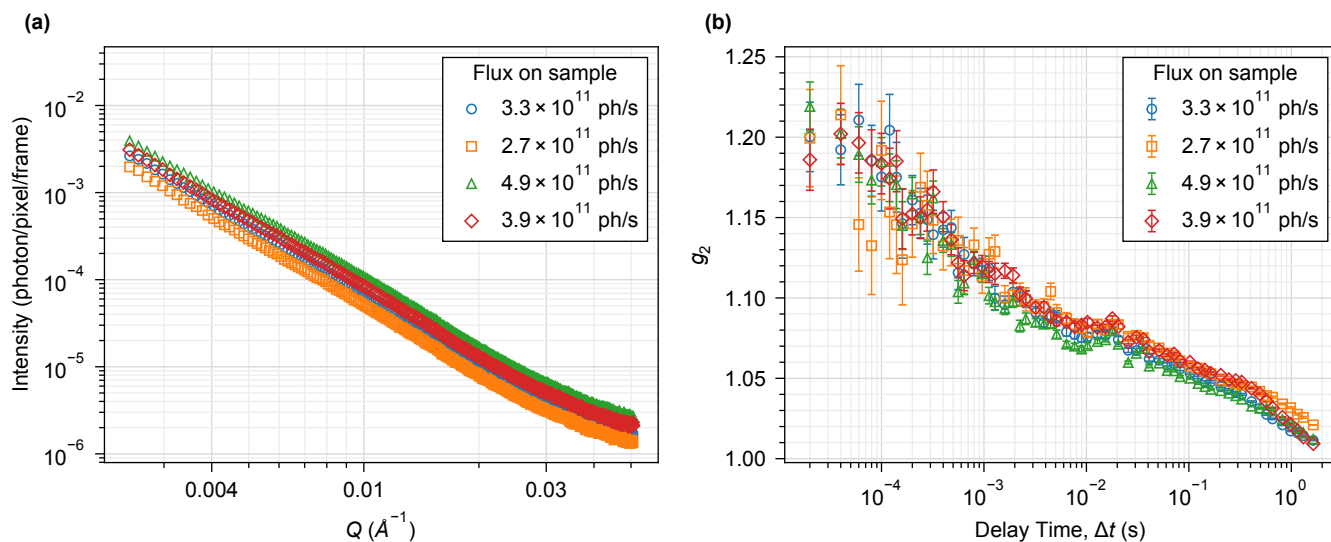


Figure S5: Flux-dependence control for SAXS and SA-XPCS. (a) SAXS profiles and (b) intensity autocorrelation functions from a 20 mg mL⁻¹ (VPAVG)₃₀ sample at incident fluxes of 2.7×10^{11} – 4.9×10^{11} photons s⁻¹. The Q range shown in (a), 0.0024–0.051 Å⁻¹, covers the Q range of Figure 3a of the main text. Intensities in (a) are detector counts per pixel per frame, not on the absolute scale of Figure 3a. Panel (b) shows the lowest Q bin of this control configuration, $Q = 0.00304$ Å⁻¹ ($2\pi/Q \approx 206$ nm). The legend orders the four conditions by nominal attenuator setting, which is not monotonic in flux. No systematic flux-dependent change is resolved over this range, which substantially exceeds the flux used in the main experiment.

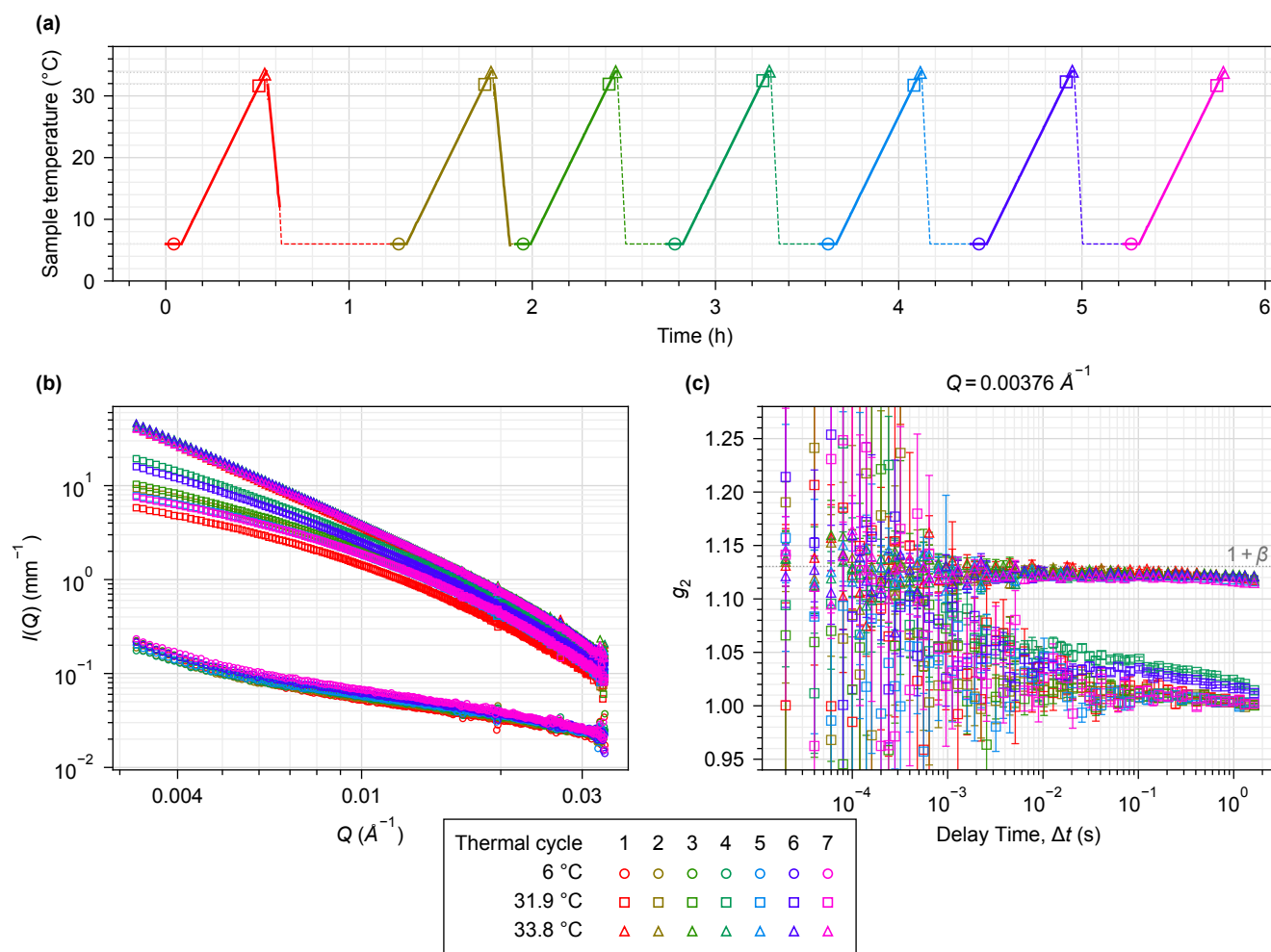


Figure S6: Thermal-cycling repeatability of the SA-XPCS sample: one aliquot—a separate loading from the isothermal experiment of Figure 3 of the main text—through seven consecutive cycles. (a) Sample temperature over the 5.8 h sequence. Solid curves are measured: the stage RTD was read once per 2 s acquisition. No acquisitions were taken while the sample cooled, so no temperature was recorded over those intervals; the dashed curves there are not data but the profile projected from the control protocol, which was fixed at 10 °C min⁻¹ cooling until 6 °C followed by a hold at 6 °C until the next ramp. Open symbols mark, at the time and temperature at which it was taken, every acquisition group that is plotted in (b) and (c), so each lower-panel curve can be traced to its point in the thermal history. (b) Buffer-subtracted SAXS on an absolute scale at 6 °C (circles, 32–41 surviving acquisitions averaged; see Section 6.2), 31.9 °C (squares, acquisitions 241–250) and 33.8 °C (triangles, acquisitions 261–270). Each group is placed on the absolute scale with the mean ion-chamber readings of the acquisitions it contains, and the averaged buffer (69 surviving of 89 acquisitions, D0077 and D0080) is subtracted with no empirical scale factor, exactly as in Figure 3a. (c) Correlation functions at the lowest Q bin for the two high-temperature states; the dotted line is $1 + \beta$ with β the contrast measured in Figure S4. Error bars are the measured pointwise g_2 uncertainties; the y axis is capped so that a few short-delay caps clip. One key serves all three panels: its columns are the seven thermal cycles, numbered along the top and colored on the same scale Figure 3 of the main text uses for elapsed time, and its rows are the three temperature states, labeled at the left and distinguished by marker shape.

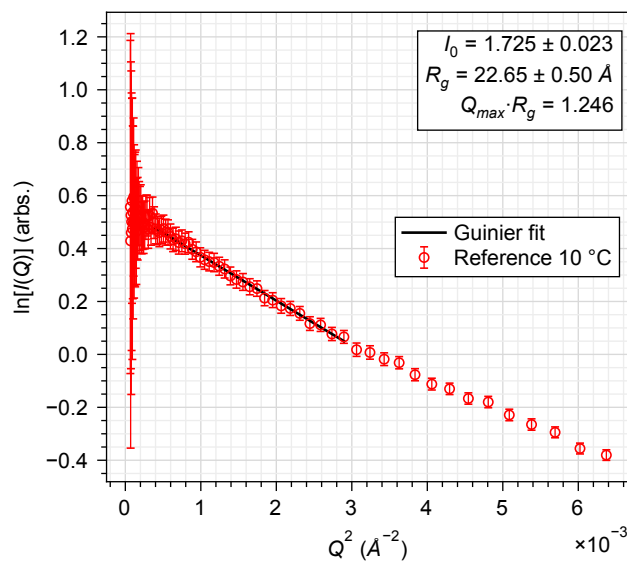


Figure S7: Guinier analysis of the Reference aliquot at 10 °C. Weighted linear fit of $\ln I(Q)$ against Q^2 over $0.020 \leq Q \leq 0.055 \text{ \AA}^{-1}$ (so $Q_{\max} R_g = 1.246$), giving an apparent $R_g = 22.65 \pm 0.50 \text{ \AA}$ and $I_0 = 1.725 \pm 0.023$ in the arbitrary intensity units of the reduced 12-ID-B profiles. Points are plotted over the wider range $0.005 \leq Q \leq 0.080 \text{ \AA}^{-1}$ for context; error bars are the propagated intensity uncertainties, σ_I/I .

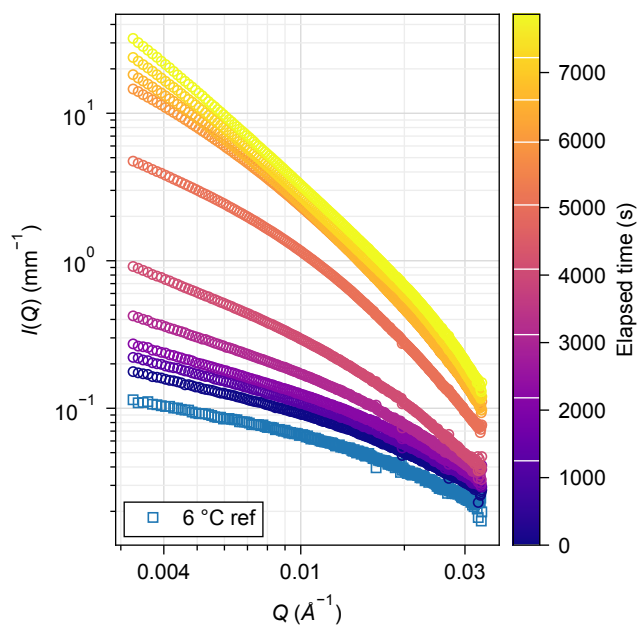


Figure S8: Complete isothermal SAXS evolution of (VPAVG)₃₀ at 30 °C, buffer-subtracted and on the same absolute scale as Figure 3a of the main text, using the same buffer measurement. The 6 °C reference and profile at the start of the hold are followed by profiles from 1249 to 7863 s. White marks on the color bar give the elapsed time of each measured profile. Colors here are scaled linearly over the full 0–7863 s range and therefore do not match the discrete colors used for the same times in Figure 3; read times from the color bar. The subset from 5039 to 7863 s corresponds to the SAXS and SA-XPCS data analyzed in Figure 3 of the main text.

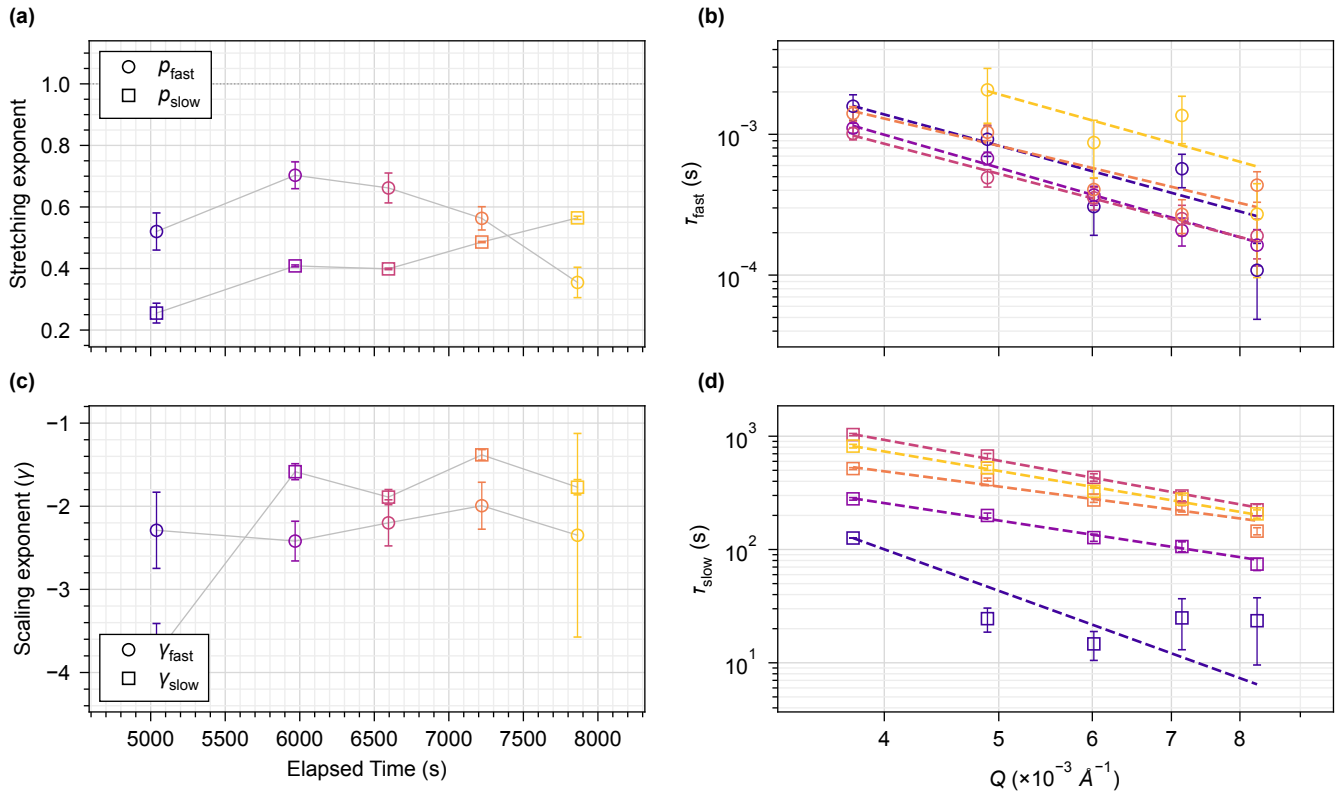


Figure S9: Additional parameters from global two-mode fits. Colors denote elapsed time as in Figure 3 of the main text (5039, 5969, 6595, 7221 and 7863 s, dark to light). (a) Stretching exponents shared across the five Q bins at each elapsed time. (b) Effective fast relaxation time versus Q . (c) Fitted scaling exponents γ_{fast} and γ_{slow} . (d) Effective slow relaxation time versus Q . Dashed lines in panels (b,d) are power-law fits $\tau \propto Q^\gamma$. At the final elapsed time the fast amplitude at the lowest Q bin refines to $f = 0.000 \pm 0.003$, so no fast mode is detected there and τ_{fast} is undefined; that point is omitted from (b) and from its power-law fit, which is why the final γ_{fast} has a larger uncertainty. The slow-mode quantities are extrapolations beyond the accessible delay range; γ_{slow} at the earliest elapsed time, -3.78 ± 0.37 , lies well outside the -1.4 to -1.9 spanned by the other four, and is the least constrained of the set because the slow mode carries its smallest amplitude there. Grey lines connect points belonging to the same parameter and are guides to the eye; the dotted line in (a) marks $p = 1$, the simple-exponential limit. Error bars are fit uncertainties.

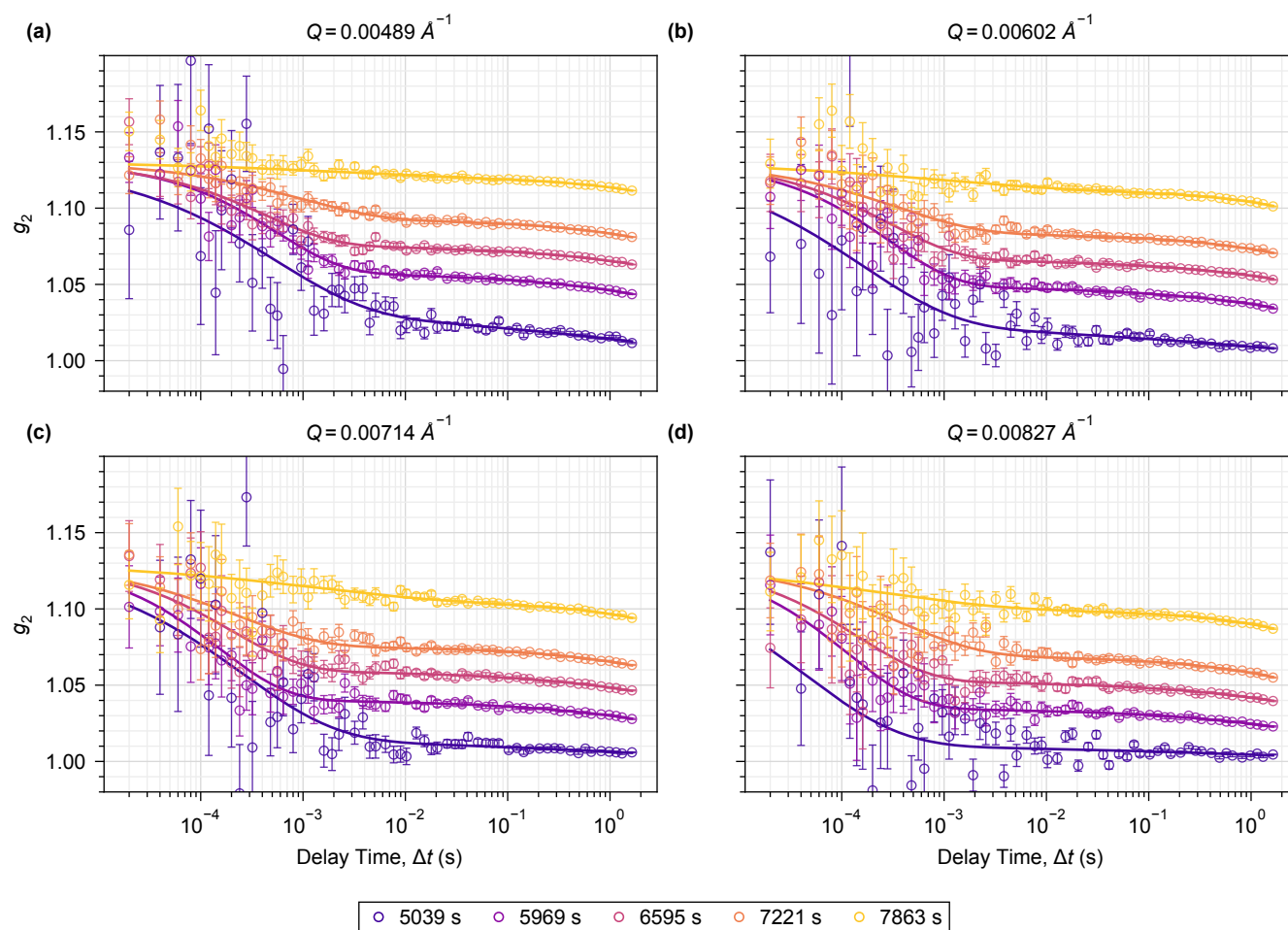


Figure S10: Expansion of Figure 3b of the main text onto the four higher Q bins. Panels show (a) $Q = 0.00489$, (b) 0.00602 , (c) 0.00714 , and (d) $0.00827 \text{ \AA}^{-1}$. Colors denote the same elapsed times as Figure 3b,c of the main text. Solid curves are the same global two-mode fits that produce Figure 3c and Figure S9: at each elapsed time all five Q bins are fitted simultaneously with p_{fast} and p_{slow} shared across Q and β fixed at the measured instrumental contrast, and the curve drawn in each panel is that global solution evaluated at that panel's Q . Error bars are the measured pointwise g_2 uncertainties. The y axis is capped at 1.20; two of 1220 plotted points and four error-bar caps, all at the shortest and noisiest delays, fall above it.

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